

Unit 3: Cells

Complete Study Notes – Grade 9 Biology

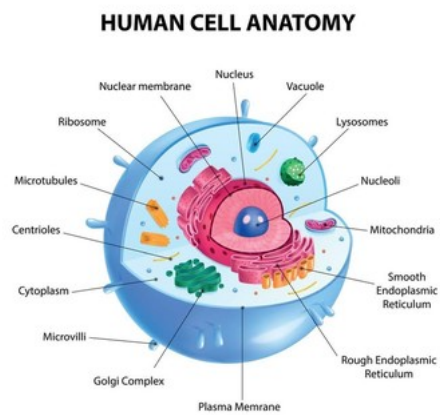


Figure 1: General structure of a eukaryotic cell showing major organelles.

© Acadia. All rights reserved.

3.1 What is a Cell?

A **cell** is the smallest and basic unit of all living things. All living organisms are composed of cells.

Unicellular Organisms:

Some living things consist of a single cell and are called **unicellular organisms**. Examples include bacteria, amoeba, and paramecium. Most unicellular organisms are invisible to the naked eye, but there are exceptions – a chicken egg cell is a single cell that can be seen without a microscope.

Multicellular Organisms:

Other living things, such as plants and animals (including humans), consist of many cells and are called **multicellular organisms**. The body of multicellular organisms is composed of many cells of specialized types that work cooperatively.

Example: In the human body, every action and every thought are possible because of processes that occur at the cellular level. Muscle cells work together for movement, nerve cells gather information, and brain cells interpret messages and coordinate actions.

Interesting Fact: An ostrich egg is a single cell about 6 inches in diameter, weighing about 1.36 kg. Nerve cells in the human body can stretch several feet long, and nerve cells in giant squid can be more than 30 feet long!

3.2 Cell Theory

The cell theory is the unifying foundation of cell biology. It was developed through the contributions of several scientists.

Robert Hooke (1665):

An English scientist, Robert Hooke, was the first to observe cells and to use the word 'cell' for structures in living organisms. He did this after examining thin slices of plant material (cork). He was surprised by the regular appearance of the structure – 'pore-like' regular structures that Hooke called 'cells' (Latin: *cellulae*, meaning 'small rooms'). Each cell appeared to be an empty box surrounded by a wall. Hooke had discovered and described a cell, without realizing that it is the fundamental unit of all living things.

Anton van Leeuwenhoek (1674):

A Dutch microbiologist, Anton van Leeuwenhoek, used his lenses to build numerous microscopes, some with magnifications approaching 300×. In 1674, he observed protozoa for the first time, and several years later he observed bacteria. He called them 'very little animalcules' and isolated them from rainwater, pond water, well water, and the human mouth and intestine. Leeuwenhoek is universally acknowledged as the **father of microbiology**.

Matthias Schleiden (1838):

A German botanist, Matthias Schleiden, stated that **all plants are composed of cells**.

Theodor Schwann (1838–1839):

A German zoologist, Theodor Schwann, concluded that **all animals are composed of cells**. He and Schleiden are credited with formulating the early cell theory.

Rudolf Virchow (1855):

A German physician, Rudolf Virchow, proposed that **all cells arise only from pre-existing cells** (*Omnis cellula e cellula*).

The Modern Cell Theory:

- All living organisms are composed of one or more cells.
- The cell is the smallest unit of life.
- All cells arise only by the division of pre-existing cells.

Common Features of All Cells:

- Every living cell is surrounded by a **membrane**, which separates the cell contents from everything outside.
- Cells contain **genetic material** (DNA) which stores all the instructions needed for the cell's activities.
- Many of these activities are chemical reactions, catalysed by **enzymes** produced inside the cell.
- Cells have their own **energy release system** that powers all of the cell's activities.

3.3 Cell Structure and Function

Cells have different structures which have different functions. In multicellular organisms, some cells have the same shape and same function, but other cells have different shapes and different functions.

Example: In the human body, the shape and function of nerve cells are different from that of muscle cells. Nerve cells are long and branched to transmit signals, while muscle cells are elongated and contain many mitochondria for energy production during contraction.

Major Structures Common to All Cells:

- **Nucleus** – Contains genetic material and controls cell activities.
- **Cytoplasm** – A thick fluid-filled region containing cell organelles where chemical reactions occur.
- **Cell (Plasma) Membrane** – A selectively permeable structure that controls what enters and leaves the cell.

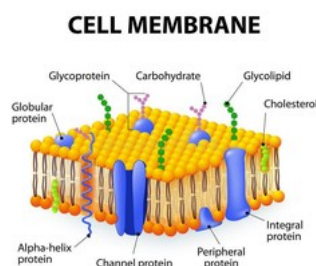


Figure 2: The cell membrane – a selectively permeable barrier.

Organelles in the Cytoplasm:

An **organelle** is a distinct structure within a cell having a specific function. Some organelles have membranes, while others do not.

Nucleus – The largest organelle, surrounded by a double membrane. Contains chromosomes (DNA + proteins). Controls cell activities and cell division. Contains the nucleolus – site of ribosome assembly.

Ribosomes – Small structures found in all cells. Function: **protein synthesis**. Found freely in the cytoplasm or attached to the rough endoplasmic reticulum.

Endoplasmic Reticulum (ER) – A complex system of interconnected double membranes. Contains fluid-filled spaces for material transport. Two types: **Rough ER** (with ribosomes – protein synthesis and storage) and **Smooth ER** (without ribosomes – lipid synthesis).

Golgi Apparatus – A smooth, concave, membranous structure. Functions in **collection, packaging, and distribution** of molecules (proteins and lipids) within or outside the cell. Named after Italian scientist Camillo Golgi (1898).

Vesicles – Small stack-like collections of membranous sacs. Store, transport, and ship materials (e.g., proteins from rough ER to Golgi bodies).

Mitochondria – Relatively large organelles with a double membrane. The inner membrane forms folds called **cristae** where some chemical reactions of respiration occur. '**Powerhouse of the cell**' – site of energy (ATP) production. Muscle cells contain thousands of mitochondria.

Lysosomes – Membrane-bound packages of digestive enzymes. Function: **digestion** – break down bacteria, foreign particles, and damaged organelles within the cell.

Vacuoles – Organelles consisting of a single membrane with fluid inside. **Plant cells** have large permanent vacuoles (occupy more than half the cell volume). **Animal cells** have small vacuoles. Types: **Food vacuoles** (digestion) and **contractile vacuoles** (expel excess water in unicellular protists).

Chloroplasts – Double membrane-bound organelles found in green plant cells (mainly in leaves). Site of **photosynthesis**. Contain **chlorophyll** (green pigment) that traps sunlight energy to synthesise carbohydrates.

Centrioles – Small cylindrical structures in animal cells. Anchor microtubules during cell division.

Cilia – Short, hair-like structures on the surface of some cells. Used for **locomotion** or creating fluid currents. Found in large numbers on the lining of the air tubes in the lungs.

Flagella – Long, thread-like structures present in smaller numbers. Extend from the plasma membrane and enable **movement** of the entire cell. Example: sperm cells have a flagellum for swimming.

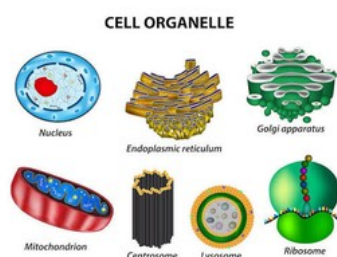


Figure 4: Key organelles – endoplasmic reticulum, Golgi apparatus, and mitochondria.

3.4 Types of Cells

Based on cell structure and cellular organisation, cells are grouped into two types: **Prokaryotic cells** and **Eukaryotic cells**.

Prokaryotic Cells:

- Usually much smaller than eukaryotic cells (1–10 µm in size).
- Have a much simpler structure.
- **No membrane-bound organelles** or a nucleus.
- Genetic material (DNA) is not enclosed in a nucleus (nucleoid region).
- Examples: **Bacteria** and **Archaea**.
- Cell wall is present (made of peptidoglycan in bacteria).

Eukaryotic Cells:

- Usually larger than prokaryotic cells (10–100 µm in size).
- Have a **true nucleus** (Greek: *eu* = true, *karyon* = kernel/nucleus).
- Have **membrane-bound organelles** (ER, Golgi, mitochondria, etc.).
- Usually **multicellular** (though some protists are unicellular).
- Examples: **Plants, animals, fungi, and protists**.

Comparison Table – Prokaryotic vs Eukaryotic Cells:

Feature	Prokaryotic Cells	Eukaryotic Cells
Nucleus	Absent (nucleoid region)	Present (true nucleus)
Membrane-bound organelles	Absent	Present (ER, Golgi, mitochondria)
Ribosomes	Smaller	Larger
Chromosomes	DNA not associated with proteins; circular	DNA associated with proteins; linear
Cell wall	Present (peptidoglycan)	Present in plants (cellulose), fungi (chitin); absent in animals
Cilia/Flagella	Some have flagella (different structure)	Some have cilia/flagella (9+2 structure)
Size	1–10 µm	10–100 µm
Examples	Bacteria, Archaea	Plants, Animals, Fungi, Protists

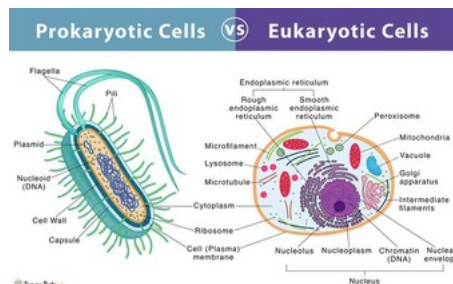


Figure 5: Comparison of prokaryotic and eukaryotic cells.

3.5 Animal and Plant Cells

Animal and plant cells have certain features in common, but they also have important differences.

Common Features (Both Plant and Animal Cells):

- Cell membrane
- Nucleus
- Cytoplasm
- Mitochondria
- Golgi apparatus

- Endoplasmic reticulum
- Ribosomes

Features Found Only in Plant Cells:

- **Cell wall** – Rigid structure made of cellulose; provides support and shape.
- **Chloroplasts** – Contain chlorophyll for photosynthesis.
- **Large permanent vacuole** – Contains cell sap; helps regulate osmosis and maintain turgor pressure.

Features Found Only in Animal Cells:

- **Centrioles** – Involved in cell division.
- **Lysosomes** – Contain digestive enzymes (some plant cells also have them, but they are more prominent in animals).

Additional Differences:

- Plant cells are usually **larger** and more easily visible under a light microscope.
- Plant cells have a **fixed shape** due to the cell wall; animal cells have a **flexible shape**.
- Plant cells store food as **starch**; animal cells store food as **glycogen**.

Example – Vacuoles: The plant vacuole is surrounded by a membrane and contains a solution of pigments, enzymes, sugars, mineral salts, oxygen, and carbon dioxide. Vacuoles help regulate the osmotic properties of cells (the flow of water inwards and outwards). In animal cells, vacuoles are small and may be used for storage or digestion.

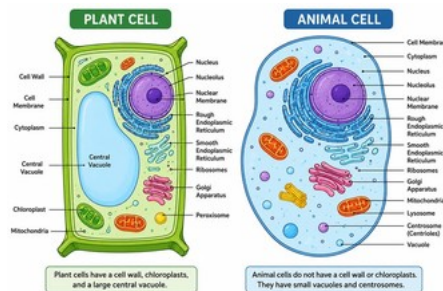


Figure 6: Differences between plant and animal cells.

3.6 Observing Cells Under a Microscope

A microscope is an instrument that can be used to observe small objects, even cells. The image of an object is magnified through at least one lens in the microscope. This lens bends light toward the eye and makes an object appear larger than it actually is.

Observing Plant Cells (Onion Epidermis):

1. Using forceps, peel a piece of epidermal tissue from the inner side of an onion bulb.
2. Place the epidermal tissue on a glass microscope slide.
3. Using a scalpel, cut out a 1 cm square of tissue and arrange it in the center of the slide.
4. Add two to three drops of **iodine solution** for staining (stains the nucleus and makes it more visible).
5. Using forceps, support a coverslip with one edge resting near the onion tissue, at an angle of about 45°.
6. Gently lower the coverslip over the onion tissue, avoiding air bubbles.

7. Leave the slide for about 5 minutes to allow the iodine stain to react with the specimen.
8. Place the slide on the microscope stage, select the lowest power objective, and focus.
9. Increase magnification using other objective lenses.
10. Make a large drawing of one cell and label: **cell wall, cell membrane, cytoplasm, nucleus.**

Observing Animal Cells (Cheek Epithelial Cells):

1. Rinse your mouth with water to remove food particles.
2. Take a cotton bud from a freshly opened pack.
3. Rub the cotton bud lightly on the inside of your cheek and gums to collect some cheek cells in saliva.
4. Rub the cotton bud onto the center of a clean microscope slide to leave a sample of saliva.
5. Add two to three drops of **methylene blue dye** for staining (stains the nucleus).
6. Using forceps, support a coverslip with one edge resting near the cheek cell sample, at an angle of about 45°.
7. Gently lower the coverslip over the tissue, avoiding air bubbles.
8. Leave the slide for a few minutes.
9. Place the slide on the microscope stage, select the lowest power objective, and focus.
10. Increase magnification using other objective lenses.
11. Make a drawing of one cell and label: **cell membrane, cytoplasm, nucleus.**
12. Place used slides in laboratory disinfectant or autoclaving before washing (safety precaution).

Safety Note: This technique involves studying epithelial cells taken from the inside of the cheek. There is a very small risk of transmitting pathogens (e.g., HIV). Appropriate precautions should be applied to treat contaminated items.

3.7 The Cell and Its Environment

The cell membrane is a **selectively permeable** structure that controls what enters and leaves the cell. Materials can pass through the cell membrane either **passively** (without energy) or **actively** (using energy).

3.7.1 Passive Transport

Passive transport is the movement of substances **down a concentration gradient** – from an area of high concentration to an area of lower concentration – without the need for input of energy.

Think of passive transport like rolling a ball **downhill** – you don't need to use any energy.

Diffusion:

Diffusion is the net movement of molecules or ions from a region of high concentration to a region of low concentration.

Examples of Diffusion:

- The smell of incense stick spreading through a room.
- Tea bag dipping in a glass of hot water – the tea diffuses throughout the water.
- Oxygen diffusing from the alveoli into the blood in the lungs.

- Carbon dioxide diffusing from the blood into the alveoli.

Factors Affecting the Rate of Diffusion:

- **Concentration gradient** – The higher the difference in concentration, the faster the diffusion rate.
- **Temperature** – Increase in temperature increases kinetic energy, so molecules move faster and diffuse faster.
- **Mass of molecule** – Heavier molecules diffuse more slowly; lighter molecules diffuse faster.
- **Distance travelled** – The farther the distance, the slower the diffusion rate.
- **Surface area** – The greater the surface area, the faster the total diffusion. Cells in the kidney and intestine have microvilli to increase surface area for absorption.

Osmosis:

Osmosis is a special case of diffusion – it is the diffusion of water molecules from a region of higher water concentration (dilute solution) to a region of lower water concentration (concentrated solution) across a selectively permeable membrane.

Osmosis in Plant Cells:

- In a **hypotonic solution** (dilute, high water concentration): Water enters the cell by osmosis. The vacuole expands, pressing against the cell wall. The plant becomes **turgid** (rigid). This is essential for plants to maintain their upright posture.
- In a **hypertonic solution** (concentrated, low water concentration): Water leaves the cell by osmosis. The plasma membrane shrinks and pulls away from the cell wall (**plasmolysis**). The plant **wilts** and may die.
- In an **isotonic solution** (same concentration): No net movement of water. The cell remains unchanged.

Osmosis in Animal Cells:

- In a **hypotonic solution** (dilute): Water enters the cell by osmosis. Animal cells have no cell wall, so the cell swells and may **burst** (hemolysis – in red blood cells).
- In a **hypertonic solution** (concentrated): Water leaves the cell by osmosis. The cell shrinks (**crenation**).
- In an **isotonic solution**: No net movement of water. The cell remains normal.

Real-World Applications of Osmosis:

- • **Food preservation:** Salting meat and fish draws water out of bacteria, preventing growth.
- • **Feeling thirsty after eating salty food:** Salt increases the concentration of solutes in the blood, causing water to move out of cells and creating thirst.
- • **Swelling of seeds/beans in water:** Water enters the seeds by osmosis, causing them to swell and germinate.
- • **Potato in pure water:** The potato swells as water enters by osmosis.

Contractile Vacuoles in Amoeba:

Unicellular protozoans such as amoeba live in freshwater (hypotonic environment). Water enters the cell by osmosis. **Contractile vacuoles** pump excess water out of the cell, preventing it from bursting. This is an example of osmoregulation.

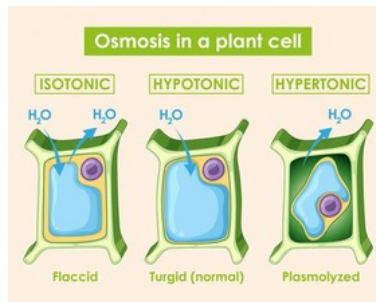


Figure 7: Osmosis – diffusion of water across a selectively permeable membrane.

3.7.2 Active Transport

Active transport is the movement of molecules or ions across the cell membrane **against a concentration gradient** (from low concentration to high concentration) using energy (ATP) from respiration.

Think of active transport like rolling a ball **uphill** – you need to use energy.

Examples of Active Transport:

- • **Sodium-potassium pump:** Animal nerve cells pump sodium ions out of the cell and potassium ions into the cell, maintaining concentration differences essential for nerve function.
- • **Ion uptake by root hairs:** Plants absorb mineral ions (e.g., nitrate, phosphate) from the soil against a concentration gradient.
- • **Glucose uptake by epithelial cells of villi:** Glucose is absorbed from the intestine into the blood, even when the concentration of glucose in the blood is already high.

Comparison of Passive and Active Transport:

Feature	Passive Transport	Active Transport
Direction	High → Low concentration	Low → High concentration
Energy required	No	Yes (ATP)
Examples	Diffusion, Osmosis	Sodium-potassium pump, Ion uptake
Cell membrane	Selectively permeable	Selectively permeable

3.8 Levels of Biological Organization

The organisation of the biological world is **hierarchical** – each level builds on the level below it.

Levels of Biological Organization (from simplest to most complex):

- 1. Atoms and Molecules** – Atoms join to form molecules (e.g., DNA, proteins).
- 2. Organelles** – Complex biological molecules are assembled into tiny structures (e.g., mitochondria, chloroplasts).
- 3. Cells** – The smallest unit of life (e.g., nerve cell, muscle cell, red blood cell).
- 4. Tissues** – Groups of similar cells that act as a functional unit (e.g., nerve tissue, muscle tissue, xylem tissue in plants).
- 5. Organs** – Body structures composed of several different tissues that work together (e.g., heart, brain, stomach, leaves, roots).

6. Organ Systems – Groups of organs that work together to perform a major function (e.g., digestive system, nervous system, circulatory system).

7. Organism – An individual living thing (e.g., a human, a tree, a fish).

8. Population – All members of the same species living in a specific area.

9. Community – All the different populations of organisms living in a specific area.

10. Ecosystem – The community of organisms plus the abiotic (non-living) factors in their environment.

11. Biosphere – The part of the Earth that supports life; the sum of all ecosystems.

Examples in Plants:

- **Tissues:** Epidermis, palisade tissue, spongy tissue, xylem, phloem.
- **Organs:** Root, stem, leaf, flower.
- **Organ Systems:** Shoot system (above ground), root system (below ground).

Examples in Animals:

- **Tissues:** Epithelial tissue, muscle tissue, nerve tissue, connective tissue.
- **Organs:** Brain, heart, lungs, stomach, eyes.
- **Organ Systems:** Digestive system, respiratory system, circulatory system, nervous system.

Example – The Digestive System: The digestive system consists of the mouth, oesophagus, stomach, small intestine, large intestine, liver, pancreas, and gall bladder. Each organ is composed of tissues (e.g., muscle tissue, epithelial tissue). The tissues are composed of cells. The digestive system works with other systems (e.g., circulatory system) to keep the organism healthy.

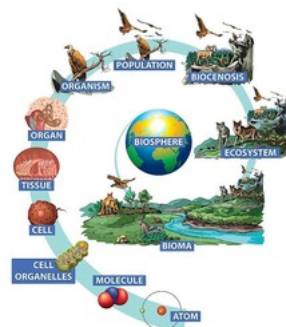


Figure 8: Levels of biological organisation – from cells to biosphere.

Unit 3 Summary

- **Cell theory:** All organisms are composed of one or more cells. The cell is the smallest unit of life. All cells arise from pre-existing cells.
- **Prokaryotic vs Eukaryotic:** Prokaryotic cells have no nucleus or membrane-bound organelles (e.g., bacteria). Eukaryotic cells have a nucleus and membrane-bound organelles (e.g., plants, animals, fungi, protists).
- **Key organelles:** Nucleus (control centre), ribosomes (protein synthesis), ER (protein/lipid transport), Golgi (packaging), mitochondria (energy production), lysosomes (digestion), vacuoles (storage), chloroplasts (photosynthesis).
- **Plant vs Animal cells:** Plant cells have a cell wall, chloroplasts, and large permanent vacuoles. Animal cells have centrioles and lysosomes.
- **Diffusion:** Movement of particles from high to low concentration. Factors: concentration gradient, temperature, molecule mass, distance, surface area.

- **Osmosis:** Diffusion of water across a selectively permeable membrane. In hypotonic solution, water enters the cell. In hypertonic solution, water leaves the cell.
- **Active transport:** Movement of particles against a concentration gradient using energy (ATP).
- **Levels of organisation:** Atoms → Molecules → Organelles → Cells → Tissues → Organs → Organ Systems → Organism → Population → Community → Ecosystem → Biosphere.

End of Unit 3 – Cells

© Acadia. All rights reserved.